

MARIA INSA

DATA ANALYTICS & VISUALISATION ENGINEER | DATA SCIENCE & DECISION-SUPPORT SYSTEMS
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- A **Data Analytics & Visualisation Engineer with 7+ years of experience** transforming complex operational and time-series data into analytical reporting, operational insights, and decision-support tools. Experienced building analytical data pipelines, validation workflows, dashboards, monitoring systems, and data-driven applications using Python, SQL, R, and cloud-based platforms.
- Worked across engineering, product, research, and stakeholder-facing environments to translate business and operational challenges into actionable analytical outputs, with a strong focus on data quality, usability, communication, and informed decision-making.

Core Expertise

- Data Analytics & Data Science: KPI reporting, operational analytics, statistical analysis, time-series analysis, anomaly detection, trend analysis, exploratory analysis, actionable insights
- Data Visualisation & Reporting: Power BI, Tableau, Grafana, Plotly, interactive dashboards, self-service reporting, data storytelling, visualisation standards and best practices
- Data Engineering & Cloud Platforms: Azure Data Explorer (ADX), Azure Data Lake, ETL pipelines, structured and analysis-ready datasets, cloud-based data workflows, data validation frameworks
- Programming & Tools: Python (pandas), SQL, R, APIs, Git, reproducible workflows, analytical tooling
- Business & Analytical Problem Solving: Translating business requirements into analytical solutions, reporting logic, root-cause investigations, operational monitoring, decision-support systems
- Communication & Collaboration: Stakeholder engagement, requirements gathering, presenting complex analysis clearly, cross-functional collaboration

Relevant Work Experience

Data Visualisation & Analytics Engineer | Intelligent Growth Solutions Ltd

Edinburgh, UK | Feb 2023 - Dec 2025 | PROJECT: "[CROP HEIGHT TRACKER](#) and other".

- Analysed large-scale operational and time-series datasets to identify trends, support operational improvements, and inform data-driven decision-making
- Built and maintained analytical data pipelines, validation frameworks, and reporting workflows using Python, SQL, and Azure Data Explorer (ADX), improving data quality, reliability, and traceability
- Developed analytical tools for anomaly detection, root-cause investigation, operational failure analysis, and system performance monitoring
- Designed and delivered operational dashboards and analytical applications used to monitor KPIs, investigate failures, explore system behaviour, and support operational decision-making
- Translated business and operational challenges into analytical outputs, KPI reporting, and reporting solutions in collaboration with engineering, product, and scientific teams
- Automated reporting, monitoring, and validation processes, improving consistency, scalability, and reproducibility of analysis

Technologies: Python, R, SQL, Azure Data Explorer (ADX), Grafana, Plotly.js, Angular, Git

Technology & Research Fellow | Glasgow Caledonian University & Centre for Theology and Inquiry

Remote (UK/USA) | Feb 2021 - Apr 2022 | PROJECT: [“VIDEOGAME FOR NEURODIVERGENT PEOPLE”](#).

- Led a £300k grant-funded project, developing a data-informed platform for end users from concept-delivery
- Designed and evaluated data-driven interaction systems, combining quantitative and qualitative analysis to evaluate user behaviour and inform system design decisions
- Worked directly with stakeholders and end users to gather requirements, analyse insights, and translate findings into structured outputs guiding platform development

UX Designer (Data Visualisation) | Lupovis

Remote (UK) | Oct 2020 - Feb 2021

- Designed dashboards and visualisation concepts in collaboration with clients to support cybersecurity analytics, monitoring, and operational decision-making
- Translated complex datasets and stakeholder requirements into clear interface designs for system monitoring and decision support
- Contributed to MVP delivery within a cross-functional and client-facing environment, supporting funding outcomes

UX Designer (Data & Analytics Interfaces) | Glasgow Caledonian University

Remote, UK | May 2020 - Sep 2020 | PROJECT: [“3D VIRTUAL TOUR OF GCU – GLASGOW CAMPUS”](#).

- Delivered a web-based platform used by 20K+ users, working with stakeholders to translate user insights and organisational needs into a functional product
- Conducted user research and iterative analysis to translate stakeholder and user requirements into structured product outputs
- Collaborated with cross-functional teams to align outputs with user and organisational requirements

Product Designer (Data Visualisation) | Geckotech Solutions Ltd

Edinburgh, UK | Jun 2017 - Aug 2019 | PROJECT: [“IMMERSIVE SYSTEM FOR STRUCTURAL INSPECTION”](#).

- Contributed to development of a data pipeline (collection, processing, visualisation) for infrastructure analysis
- Designed visual interfaces to support interpretation and analysis of complex engineering datasets
- Delivered a data acquisition system enabling engineers to capture and generate 3D infrastructure models

Education

PhD, Human-Computer Interaction, Data Vis & AI - Glasgow Caledonian University, UK (2018-2023)

BEng (Hons), Product Design & Development - Polytechnic University of Catalonia, Spain (2013-2017)

Other Work Experience

Teaching Assistant | Glasgow Caledonian University, Remote, UK | Oct 2019 - Jun 2020

Hackathon & Competitions - EU Datathon by EU Data Viz, [Cop-26 Climate Data Challenge](#) by Met Office, [Coronahack by Mindstream AI](#) and [Formula 1 Student Team](#) by Institution of Mechanical Engineers

Skills

- Visualisation & Reporting: Power BI, Tableau, Grafana, Plotly, Holoviews, D3
- Analysis & Programming: Python, R, SQL, statistical analysis, APIs, Git, data pipelines
- Data Platforms: Azure (Data Explorer, Data Lake), AWS
- Tech & Web: HTML, CSS, JavaScript, Angular
- Systems: Windows/macOS (Proficient), Linux (Advanced)
- Languages: English (Fluent), Spanish/Catalan (Native), Flamenco (Native expressive language)